

ABDUL FATAH AMRULLOH

Telecommunications Engineering Student

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📍 Indonesia



SUMMARY

A Telecommunications Engineering graduate from Telkom University with a strong foundation in Internet of Things (IoT), Adaptive Networks, and intelligent network simulation. Experienced in developing VANET simulations, IoT-based applications, network analysis, and full-stack web development through various academic and professional projects. Currently working as a Technical Support Broadcasting professional at Trans TV, responsible for broadcast operations, MCR system monitoring, broadcast troubleshooting, and developing web-based shift scheduling systems. Proficient in system development, problem-solving, cross-functional collaboration with a strong ability to adapt and deliver innovative solutions in dynamic environments

EDUCATION

Bachelor's Degree in Telecommunication Engineering

Telkom University

📅 08/2021 - 06/2025 📍 Indonesia

CERTIFICATIONS

CCNA V7: Introduction to Networks

Cisco Certified Network Associate (CCNA) v7: Introduction to Networks

CCNA v7: Switching, Routing, and Wireless Essentials

CCNAv7: Switching, Routing, and Wireless Essentials

Microsoft Azure AI Fundamentals

Introduction to AI, machine learning, anomaly detection, computer vision, natural language processing, knowledge mining, AI challenges and risks, responsible AI

HCIA-AI V3.5 Course

Gained an understanding of artificial intelligence (AI) fundamentals and its real-world applications, Studied neural networks, machine learning models, and AI system architectures

Scaling AI in Your Organization

Explored strategies for responsibly scaling AI adoption in organizational environments, Understood the impact of AI on productivity, decision-making, and efficiency

Prompt Engineering with Azure OpenAI Service

Learned the basics of prompt engineering for large language models (LLMs), Applied prompt engineering in practical scenarios using Azure OpenAI Service

EXPERIENCE

Technical Operation Support

Televisi Transformasi Indonesia (TransTV)

📅 2025 - Present 📍 Indonesia

- Operated and monitored the Master Control Room (MCR) system to ensure broadcasts ran according to schedule
- Troubleshooted broadcast disruptions including signal loss, audio desync, and server downtime
- Managed network connectivity between MCR, studio, and editing rooms, and performed preventive maintenance
- Developed various innovative technology solutions across IoT, web and android development, AI, and broadcasting systems, focusing on automation, system integration, and improving operational performance

Intern

PT Sisterponik

📅 10/2025 - Present 📍 Indonesia

A technology company focused on IoT applications

- Designed schematics and boards with Eagle and Fritzing software, and casing design
- Conducted on-air technical troubleshooting to ensure optimal transmission quality
- Coordinated with production teams and master control for broadcast scheduling
- Verified and prepared broadcast materials according to station technical standards
- Maintained transmission system stability and reported technical incidents
- Integrated hardware and software and conducted field testing for data validation
- Supported the manufacturing process and data collection experiments in agricultural soil

Teaching Assistant

Programming Laboratory and Network Security Laboratory

📅 09/2023 - 02/2023 📍 Telkom University

Educational laboratory focused on programming and network security

- Prepared materials before laboratory sessions
- Assisted students in simulating MATLAB
- Evaluated simulation results of students

PROJECTS

Smart Poultry Monitoring System (SMAJA)

📅 2021 📍 Purwokerto, Indonesia

Designed a mobile application prototype to support digital poultry investment and monitoring activities

- Created UI/UX design with interactive investment dashboards and transaction features
- Designed user flow, navigation, and component-based prototypes
- Developed interface concepts focused on usability and user experience
- Technology: Figma (UI/UX Design), Android grid layout, Modern illustrations, Flat iconography, Component-based prototyping with interactive transitions

IoT-Based Smart Soil Monitoring System for NPK Detection

📅 2023 📍 Purwokerto, Indonesia

Developed an IoT-based monitoring system to measure and visualize real-time soil nutrient conditions

- Integrated NPK sensors with Arduino for soil nutrient data collection
- Implemented Modbus communication for sensor data transmission
- Developed a web-based dashboard for real-time monitoring
- Created a user-friendly interface for accessing soil information
- Technology: NPK Sensor, Arduino, Modbus, SIM800L, Python, C++, HTML, CSS, JavaScript, Blynk

AODV Routing Protocol Simulation on VANETs

📅 2025 📍 Purwokerto, Indonesia

Developed a VANET simulation system to analyze routing performance and visualize network communication efficiency

- Configured vehicle mobility simulation using SUMO and network simulation using OMNeT++
- Implemented AODV routing protocol for dynamic vehicle network communication
- Analyzed network performance metrics including delay, throughput, and packet delivery ratio
- Built an interactive website for real-time simulation result visualization
- Technology: SUMO, OMNeT++, C++, Python, HTML, CSS, JavaScript

Technical Support Shift Scheduling Website

📅 2025 📍 Jakarta, Indonesia

Developed an internal full-stack platform to improve operational scheduling and team coordination

- Built dynamic shift scheduling features with real-time data access
- Automated Excel-based schedule updates to minimize manual input
- Integrated crew information management including roles and contacts
- Improved communication between operational teams through centralized data
- **Technologies:** PHP, Python, MySQL, HTML, CSS, JavaScript, Figma, Excel (.xlsx)

Technical Operational Support Management System

📅 2026 📍 Jakarta, Indonesia

Developed a web-based operational management system for technical monitoring and asset management

- Created dashboards for operational activity monitoring and reporting
- Developed inventory tracking and technical documentation features
- Improved troubleshooting efficiency through centralized data management
- Designed responsive interfaces for operational environments
- **Technologies:** PHP, Python, MySQL, HTML, CSS, JavaScript

EXPERIENCE

Teaching Assistant

Programming Laboratory and Network Security Laboratory

📅 09/2023 - 02/2023 📍 Telkom University

Educational laboratory focused on programming and network security

- Prepared materials before laboratory sessions
- Assisted students in simulating Cadsoft Eagle Design PCB, Autodesk Viewer, AutoCAD, and SolidWorks
- Evaluated simulation results of students

Teaching Assistant

Programming Laboratory and Network Security Laboratory

📅 09/2023 - 02/2023 📍 Telkom University

Educational laboratory focused on programming and network security

- Prepared materials before laboratory sessions
- Assisted students in simulating Python using Visual Studio Code and Google Colab
- Evaluated simulation results of students

KEY ACHIEVEMENTS

📷 Android Live Streaming Application (SRT Broadcaster)

Developed an Android live streaming application enabling real-time low-latency video broadcasting using SRT protocol for professional streaming environments

💎 Video Decoder Dashboard & Broadcasting Network Management System

Developed an integrated broadcasting management dashboard for real-time video monitoring, multi-source stream management, and network control using MediaMTX and streaming protocols

SKILLS

Additional Skills

Python	C++	HTML	CSS
JavaScript	Node.js	IoT	
VANET	OMNeT++	NS3	
PCB design			
Cadsoft Eagle Design PCB			
Autodesk Viewer	AutoCAD		
SolidWorks			

PROJECTS

AI-Based Smart Video Cutting and Framing System

📅 2026 📍 Jakarta, Indonesia

Developed an AI-powered video processing platform to automate short-form content creation

- Integrated YouTube video processing through URL-based input
- Automated video cutting and batch processing for short clips
- Implemented AI face detection and tracking for smart framing
- Supported multiple output formats for social media platforms
- **Technologies:** Python, OpenCV, FFmpeg, YouTube API, PHP, HTML, CSS, JavaScript

Android Live Streaming Application (SRT Broadcaster)

📅 2026 📍 Jakarta, Indonesia

Developed a professional mobile live streaming application using low-latency streaming technology

- Implemented real-time video streaming from Android devices using SRT protocol
- Configured H.264 video encoding and AAC audio processing
- Added streaming controls including camera switching, preview, and status monitoring
- **Technologies:** Android, Java/Kotlin, StreamPack, SRT, H.264/AVC, AAC-LC, 4G/5G/Wi-Fi

Video Decoder Dashboard & Broadcasting Network Management System

📅 2026 📍 Jakarta, Indonesia

Developed an integrated broadcasting control dashboard for video source management and network monitoring

- Managed multiple streaming sources including Android devices, DeckLink capture cards, and SRT streams
- Implemented real-time preview, recording, and video repository management
- Developed network monitoring and device detection features.
- Integrated broadcasting protocols for professional live production environments
- **Technologies:** MediaMTX, FFmpeg, Blackmagic DeckLink SDK, SRT, RTMP, RTSP, HLS, .NET WPF, YAML Parser, ICMP Network Diagnostics

STRENGTHS



Technical Skills

Solid background in computer networks and IoT development



Interpersonal Skills

Strong communication, teamwork, problem-solving, adaptability, analytical thinking, time management, and continuous learning

LANGUAGES

Indonesian

Native



English

Proficient

